5.5 Exponential Random Variables

A continuous random variable whose probability density function is given, for some $\lambda>0$, by

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & \text{if } x\geq 0\\ 0 & \text{if } x<0\end{cases}$$

is said to be an *exponential* random variable (or, more simply, is said to be exponentially distributed) with parameter λ. The cumulative distribution function $F(a)$ of an exponential random variable is given by

$$F(a) = P\{X\leq a\} = \int_0^a \lambda e^{-\lambda x}\, dx = -e^{-\lambda x}\,|_0^a = 1-e^{-\lambda a} \qquad a\geq 0$$

Note that $F(\infty)=\int_0^\infty \lambda e^{-\lambda x}\, dx=1$, as, of course, it must. The parameter λ will now be shown to equal the reciprocal of the expected value.

Example 5a

Let X be an exponential random variable with parameter λ. Calculate (a) $E[X]$ and (b) $Var(X)$.

Solution

(a) Since the density function is given by

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & x\geq 0\\ 0 & x<0\end{cases}$$

we obtain, for $n>0$,

$$E[X^n]=\int_0^\infty x^n \lambda e^{-\lambda x}\, dx$$

Integrating by parts (with $\lambda e^{-\lambda x}=dv$ and $u=x^n$) yields

$$E[X^n] = -x^n e^{-\lambda x}\,|_0^\infty + \int_0^\infty e^{-\lambda x} n x^{n-1}\, dx = 0 + \frac{n}{\lambda}\int_0^\infty \lambda e^{-\lambda x} x^{n-1}\, dx = \frac{n}{\lambda} E[X^{n-1}]$$

Letting $n=1$ and then $n=2$ gives

$$E[X]= \frac{1}{\lambda} E[X^2]= \frac{2}{\lambda} E[X]= \frac{2}{\lambda^2}$$

(b) Hence,

$$Var(X)= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Thus, the mean of the exponential is the reciprocal of its parameter λ, and the variance is the mean squared.

In practice, the exponential distribution often arises as the distribution of the amount of time until some specific event occurs. For instance, the amount of time (starting from now) until an earthquake occurs, or until a new war breaks out, or until a telephone call you receive turns out to be a wrong number are all random variables that tend in practice to have exponential distributions. (For a theoretical explanation of this phenomenon, see **Section 4.7**.)

Example 5b

Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda=\frac{1}{10}$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait

a. more than 10 minutes;
b. between 10 and 20 minutes.

Solution

Let X denote the length of the call made by the person in the booth. Then the desired probabilities are

a. $P\{X>10\} = 1-F(10) = e^{-1} \approx .368$

b. $P\{10<X<20\} = F(20)-F(10) = e^{-1} - e^{-2} \approx .233$

We say that a nonnegative random variable X is *memoryless* if

$$P\{X>s+t \mid X>t\}=P\{X>s\} \quad \text{for all } s, t \geq 0 \tag{5.1}$$

If we think of X as being the lifetime of some instrument, **Equation (5.1)** states that the probability that the instrument survives for at least s+t hours, given that it has survived t hours, is the same as the initial probability that it survives for at least s hours. In other words, if the instrument is alive at age t, the distribution of the remaining amount of time that it survives is the same as the original lifetime distribution. (That is, it is as if the instrument does not "remember" that it has already been in use for a time t.)

Equation (5.1) is equivalent to

$$P\{X>s+t,X>t\}P\{X>t\}=P\{X>s\}$$

or

$$P\{X>s+t\}=P\{X>s\}P\{X>t\} \tag{5.2}$$

Since **Equation (5.2)** is satisfied when X is exponentially distributed (for $e^{-\lambda(s+t)}=e^{-\lambda s}e^{-\lambda t}$), it follows that exponentially distributed random variables are memoryless.

Example 5c

Consider a post office that is staffed by two clerks. Suppose that when Mr. Smith enters the system, he discovers that Ms. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Ms. Jones or Mr. Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with parameter λ, what is the probability that of the three customers, Mr. Smith is the last to leave the post office?

Solution

The answer is obtained by reasoning as follows: Consider the time at which Mr. Smith first finds a free clerk. At this point, either Ms. Jones or Mr. Brown would have just left, and the other one would still be in service. However, because the exponential is memoryless, it follows that the additional amount of time that this other person (either Ms. Jones or Mr. Brown) would still have to spend in the post office is exponentially distributed with parameter λ. That is, it is the same as if service for that person were just starting at this point. Hence, by symmetry, the probability that the remaining person finishes before Smith leaves must equal 12.

It turns out that not only is the exponential distribution memoryless, but it is also the unique distribution possessing this property. To see this, suppose that X is memoryless and let $\bar{F}(x)=P\{X>x\}$. Then, by **Equation (5.2)**,

$$\bar{F}(s+t)=\bar{F}(s)\bar{F}(t)$$

That is, $\bar{F}(\cdot)$ satisfies the functional equation

$$g(s+t)=g(s)g(t)$$

However, it turns out that the only right continuous solution of this functional equation is[†]

†One can prove **Equation (5.3)** as follows: If $g(s+t)=g(s)g(t)$, then

$$g(2n)=g(1n+1n)=g2(1n)$$

and repeating this yields $g(m/n)=gm(1/n)$. Also,

$$g(1)=g(1n+1n+\cdots+1n)=gn(1n) \quad \text{or} \quad g(1n)=(g(1))1/n$$

Hence, $g(m/n)=(g(1))m/n$, which, since g is right continuous, implies that $g(x)=(g(1))x$. Because $g(1)=(g(12))2\geq 0$, we obtain $g(x)=e-\lambda x$, where $\lambda=-\log(g(1))$.

$$g(x)=e-\lambda x \tag{5.3}$$

and, since a distribution function is always right continuous, we must have

$$\bar{F}(x)=e-\lambda x \quad \text{or} \quad F(x)=P\{X\leq x\}=1-e-\lambda x$$

which shows that X is exponentially distributed.

Example 5d

Suppose that the number of miles that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 miles. If a person desires to take a 5000-mile trip, what is the probability that he or she will be able to complete the trip without having to replace the car battery? What can be said when the distribution is not exponential?

Solution

It follows by the memoryless property of the exponential distribution that the remaining lifetime (in thousands of miles) of the battery is exponential with parameter $\lambda=110$. Hence, the desired probability is

$$P\{\text{remaining lifetime} > 5\}=1-F(5)=e-5\lambda=e-1/2\approx .607$$

However, if the lifetime distribution *F* is not exponential, then the relevant probability is

$$P\{\text{lifetime}>t+5|\text{lifetime}>t\}=1-F(t+5)1-F(t)$$

where t is the number of miles that the battery had been in use prior to the start of the trip. Therefore, if the distribution is not exponential, additional information is needed (namely, the value of t) before the desired probability can be calculated.

A variation of the exponential distribution is the distribution of a random variable that is equally likely to be either positive or negative and whose absolute value is exponentially distributed with parameter $\lambda, \lambda \geq 0$. Such a random variable is said to have a *Laplace* distribution,‡ and its density is given by

‡ It also is sometimes called the double exponential random variable.

$$f(x)=\frac{1}{2}\lambda e^{-\lambda|x|} \quad -\infty<x<\infty$$

Its distribution function is given by

$$F(x) = \begin{cases} \frac{1}{2}\int_{-\infty}^{x} \lambda e^{\lambda y}\, dy & x<0 \\ \frac{1}{2}\int_{-\infty}^{0} \lambda e^{\lambda y}\, dy + \frac{1}{2}\int_{0}^{x} \lambda e^{-\lambda y}\, dy & x>0 \end{cases} = \begin{cases} \frac{1}{2} e^{\lambda x} & x<0 \\ 1-\frac{1}{2} e^{-\lambda x} & x>0 \end{cases}$$

Example 5e

Consider again **Example 4e**, which supposes that a binary message is to be transmitted from A to B, with the value 2 being sent when the message is 1 and −2 when it is 0. However, suppose now that rather than being a standard normal random variable, the channel noise N is a Laplacian random variable with parameter $\lambda=1$. Suppose again that if R is the value received at location B, then the message is decoded as follows:

If $R \geq .5$, then 1 is concluded.

If $R<.5$, then 0 is concluded.

In this case, where the noise is Laplacian with parameter $\lambda=1$, the two types of errors will have probabilities given by

$$P\{\text{error}|\text{message 1 is sent}\} = P\{N<-1.5\} = \frac{1}{2} e^{-1.5} \approx .1116$$
$$P\{\text{error}|\text{message 0 is sent}\} = P\{N \geq 2.5\} = \frac{1}{2} e^{-2.5} \approx .041$$

On comparing this with the results of **Example 4e**, we see that the error probabilities are higher when the noise is Laplacian with $\lambda=1$ than when it is a standard normal variable.

5.5.1 Hazard Rate Functions

Consider a positive continuous random variable X that we interpret as being the lifetime of some item. Let X have distribution function F and density f. The *hazard rate* (sometimes called the *failure rate*) function $\lambda(t)$ of F is defined by

$$\lambda(t)=\frac{f(t)}{\bar{F}(t)}, \quad \text{where } \bar{F} = 1 - F$$

To interpret $\lambda(t)$, suppose that the item has survived for a time t and we desire the probability that it will not survive for an additional time *dt*. That is, consider $P\{X \in (t,t+dt)|X>t\}$. Now,

$$P\{X \in (t,t+dt)|X>t\} = \frac{P\{X \in (t,t+dt), X>t\}}{P\{X>t\}} = \frac{P\{X \in (t,t+dt)\}}{P\{X>t\}} \approx \frac{f(t)}{\bar{F}(t)}\, dt$$

Thus, $\lambda(t)$ represents the conditional probability intensity that a t-unit-old item will fail.

Suppose now that the lifetime distribution is exponential. Then, by the memoryless property, it follows that the distribution of remaining life for a t-year-old item is the same as that for a new item. Hence, $\lambda(t)$ should be constant. In fact, this checks out, since

$$\lambda(t) = f(t)\,\bar{F}(t) = \lambda e^{-\lambda t} e^{-\lambda t} = \lambda$$

Thus, the failure rate function for the exponential distribution is constant. The parameter λ is often referred to as the *rate* of the distribution.

It turns out that the failure rate function $\lambda(s)$, $s \geq 0$, uniquely determines the distribution function F. To prove this, we integrate $\lambda(s)$ from 0 to t to obtain

$$\int_0^t \lambda(s)ds = \int_0^t \frac{f(s)}{1-F(s)}\,ds = -\log(1-F(s)) \Big|_0^t = -\log(1-F(t)) + \log(1-F(0)) = -\log(1-F(t))$$

where the second equality used that $f(s) = \frac{d}{ds}F(s)$ and the final equality used that $F(0) = 0$. Solving the preceding equation for $F(t)$ gives

$$F(t) = 1 - \exp\left\{-\int_0^t \lambda(s)\,ds\right\} \tag{5.4}$$

Hence, a distribution function of a positive continuous random variable can be specified by giving its hazard rate function. For instance, if a random variable has a linear hazard rate function—that is, if

$$\lambda(t) = a + bt$$

then its distribution function is given by

$$F(t) = 1 - e^{-at - bt^2/2}$$

and differentiation yields its density, namely,

$$f(t) = (a + bt)e^{-(at + bt^2/2)} \quad t \geq 0$$

When $a = 0$, the preceding equation is known as the *Rayleigh density function*.

Example 5f

One often hears that the death rate of a person who smokes is, at each age, twice that of a nonsmoker. What does this mean? Does it mean that a nonsmoker has twice the probability of surviving a given number of years as does a smoker of the same age?

Solution

If $\lambda_s(t)$ denotes the hazard rate of a smoker of age t and $\lambda_n(t)$ that of a nonsmoker of age t, then the statement at issue is equivalent to the statement that

$$\lambda_s(t) = 2\lambda_n(t)$$

The probability that an A-year-old nonsmoker will survive until age B, $A < B$, is

$$P\{\text{A-year-old nonsmoker reaches age B}\} = P\{\text{nonsmoker's lifetime} > B \mid \text{nonsmoker's lifetime} > A\} = \frac{\bar{F}_{non}(B)}{\bar{F}_{non}(A)} = \frac{\exp\left\{-\int_0^B \lambda_n(t)\,dt\right\}}{\exp\left\{-\int_0^A \lambda_n(t)\,dt\right\}} = \exp\left\{-\int_A^B \lambda_n(t)\,dt\right\} \tag{5.14}$$

whereas the corresponding probability for a smoker is, by the same reasoning,

$$P\{\text{A-year-old smoker reaches age B}\} = \exp\left\{-\int_A^B \lambda_s(t)\,dt\right\} = \exp\left\{-2\int_A^B \lambda_n(t)\,dt\right\} = \left[\exp\left\{-\int_A^B \lambda_n(t)\,dt\right\}\right]^2$$

In other words, for two people of the same age, one of whom is a smoker and the other a nonsmoker,

the probability that the smoker survives to any given age is the *square* (not one-half) of the corresponding probability for a nonsmoker. For instance, if $\lambda_n(t)=130$, $50 \le t \le 60$, then the probability that a 50-year-old nonsmoker reaches age 60 is $e^{-1/3} \approx .7165$, whereas the corresponding probability for a smoker is $e^{-2/3} \approx .5134$.

Equation (5.4) can be used to show that only exponential random variables are memoryless. For if a random variable has a memoryless distribution then the remaining life of a s year old must be the same for all s. That is, if X is memoryless, then $\lambda(s)=c$. But, by **Equation (5.4)**, this implies that the distribution function of X is $F(t)=1-e^{-ct}$, showing that X is exponential with rate c.